

Market Basket Analysis & Business Solutions

Automated Retail Cross-Sell Strategy Report | KNIME & Excel Pipeline

1. Core Business Problems Identified

- Stagnant Average Order Value (AOV):** Customers frequently buy single or isolated items without discovering complementary products, missing clear upselling opportunities.
- Intuition-Based Bundling:** Promotional bundles and cross-sell campaigns are created based on subjective guesses rather than empirical transaction evidence, reducing promotional ROI.
- Manual Data Processing Bottleneck:** Analyzing thousands of transaction logs manually takes excessive time, making frequent automated marketing updates unsustainable.

2. How This Analysis & Workflow Solves The Problems

- Empirical Association Rule Mining:** Evaluated 7,959 transaction rows across 3,196 customers using the Apriori algorithm to identify statistically verified product affinities (Support, Confidence, Lift).
- Automated End-to-End Pipeline:** Built a KNIME workflow (CSV Reader -> GroupBy -> Association Rule Learner -> Row Filter -> Split Collection Column -> Excel Writer) that transforms raw logs into clean exports in seconds.
- Actionable Executive Output:** Formed structured string exports (MBA.xlsx) compatible with Excel for immediate implementation by store managers and digital marketers.

3. Key Analytics Findings & Rule Metrics

Items Bought Together (Antecedent)	Target (Consequent)	Confidence	Lift	Strategic Action
rolls,buns + shopping bags	whole milk	41.03%	1.251	Checkout counter pairing
other vegetables + shopping bags	whole milk	40.91%	1.248	POS Bundle discount
pastry + other vegetables	whole milk	38.89%	1.186	Aisle layout co-location
citrus fruit + other vegetables	whole milk	38.89%	1.186	Cross-category coupon
rolls,buns + sausage	whole milk	36.67%	1.118	Breakfast combo deal

4. Implementation Roadmap & Strategic Recommendations

- Digital Recommendation Triggers:** Program cart upsells on e-commerce sites so when customers add 'Other Vegetables + Citrus Fruit', 'Whole Milk' is automatically suggested at checkout.
- Store Layout Optimization:** Position high-lift staple items (Whole Milk, Bakery) in close proximity to produce sections to drive physical cross-purchases.
- Promotional Bundles:** Create packaged combo SKUs (e.g., 'Sausage + Rolls/Buns + Milk') with a 10% discount to elevate baseline Basket Size.